

Unveiling the True Power of the Latched Ring Oscillator for a Unified PUF and TRNG Architecture

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Abstract—This work presents a novel proposal for utilizing the latched ring oscillator (LRO) as a reconfigurable entropy source, outperforming the existing literature on both physical unclonable functions (PUFs) and true random number generators (TRNGs). The PUF working principle and mathematical model are proposed in this manuscript for the first time as well as its performance measured on FPGA. The proposed LRO-based PUF is 2× more compact than state-of-the-art PUFs on FPGA. The LRO TRNG architecture has been revisited, and an XOR-tree-based postprocessing technique has been introduced to increase the throughput from 0.76 up to 800 Mbit/s, paving the way for a novel class of high-throughput reconfigurable entropy sources. The results of NIST tests carried out also under supply voltage and temperature variations have demonstrated robust key extraction and secure random number generation for different applications. This comprehensive proposal aims to advance the state of the art in compact and high-throughput entropy sources, catering to the increasing demands of modern cryptographic hardware.

Index Terms—FPGA, metastability, physical unclonable function (PUF), ring oscillator (RO), true random number generator (TRNG).

I. INTRODUCTION

While cryptographic algorithms are essential to protect data, physical devices have to be secured against attacks, such as laser cutters and ion beams [1], [2], and innovative storage and authentication methods are crucial to mitigate security threats.

Physical unclonable functions (PUFs) are an excellent solution for security in lightweight systems, such as RFID and NFC, due to their unpredictable output, which adds a layer of security beyond what traditional protocols offer. PUF-based protocols enhance security by preventing tampering, probing, and cloning attacks at the device level [1], [2], [3], [4].

True random number generators (TRNGs) are fundamental cryptographic primitives, able to derive randomness from different physical phenomena, and to generate a random bitstream, with good statistical characteristics [5].

PUFs and TRNGs find application in ASIC and FPGA platforms, but the rising trend in recent research favors their development on reconfigurable and cost-effective FPGAs [1].

Various FPGA-compatible solutions exist for PUF integration. SRAM-based PUFs, such as those in [1] and [2], offer compactness and superior uniqueness due to metastable cell asymmetries and typically outperform implementations based on conventional ring oscillators (ROs). The Butterfly PUF employs a cross-coupled latch pair, which needs careful routing to achieve a robust race-condition-based bit generation. Other PUFs, such as NAND latch PUF [3] and Pico-PUF [1], [4], exhibit lower reliability and use less resources

than RO-based solutions [1]. The TERO PUF [6] exploits a transient effect, due to the mismatch between the oscillation frequency of two nominally identical RO branches, and requires extra hardware resources and postelaboration for reliable bit extraction [7]. Techniques for enhancing the reliability of metastability-based PUFs have been introduced in [2], [8], and [9], where a novel excitation sequence is exploited to achieve stable and reliable responses.

Several FPGA-based solutions exist also for TRNG integration, relying on physical phenomena, such as jitter accumulation [5] or coherent sampling (CS). RO configurations, common in TRNGs [10], demand careful delay balancing for NIST compliance. High-throughput RO-based TRNGs utilize frequency modulation or multistage-feedback RO (MSFRO) [11] techniques. Lightweight architectures allow to minimize the resources, but may result in low quality of the native bitstream, requiring more complex postprocessing [12] or feedback strategies [5].

The motivation behind utilizing a reconfigurable entropy source stems from the increasing demand for integrating both PUFs and TRNGs in cryptographic protocols. Recently, a privacy preserving mutual authentication (PPMA) scheme was proposed, which leverages the combined use of PUFs and TRNGs on both the device being authenticated and the server [9], [13]. This protocol offers several advantages over other authentication methods, including enhanced robustness against man-in-the-middle attacks. The ability to use a single entropic source that can function as either a PUF or TRNG is particularly attractive for devices with limited hardware resources [4]. Given this, there has been a growing interest in reconfigurable entropy sources in the recent literature. However, in the context of IoT security, designing a unified, reconfigurable, PUF + TRNG architecture poses many challenges, involving different tradeoffs among statistical characteristics, complexity, resources usage, and power consumption [14], [15].

An RO-based PUF + TRNG solution compatible with FPGA platforms is reported in [16]. In the circuit proposed in [10], oscillation frequencies are compared for PUF key derivation, and the XORed RO outputs are exploited to extract the random bitstream. In [16], a conventional RO PUF is repurposed as a TRNG. The solution in [17] is based on two pairs of oscillators, and the number of periods of the second oscillator across a given number of periods of the first determines the PUF key and the random sequence. The Universal TERO (UTERO) architecture reported in [18] uses TERO loops and PUF/TRNG selection logic. Despite the good performance of the PUF, the throughput of the TRNG is limited.

In this work, a novel approach exploiting the latched RO (LRO) [5] as a reconfigurable entropy source is presented. The PUF working principle and mathematical model are proposed in this manuscript for the first time as well as its performance measured on FPGA. In addition, the TRNG architecture of [5] has been revisited, and an XOR-tree-based postprocessing technique has been introduced to increase the throughput from 0.76 up to 800 Mbit/s. In order to propose a very lightweight reconfigurable entropic source (occupying only 46 Slices), in this work, we propose a novel method to extract the PUF key and a more effective way to collect random bitstream from the entropy cells to reach higher throughput (800 Mbit/s). The

Manuscript received 9 April 2024; revised 9 July 2024 and 29 July 2024; accepted 14 August 2024. Date of publication 4 September 2024; date of current version 6 December 2024. (Corresponding author: Riccardo Della Sala.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TVLSI.2024.3448503>.

Digital Object Identifier 10.1109/TVLSI.2024.3448503

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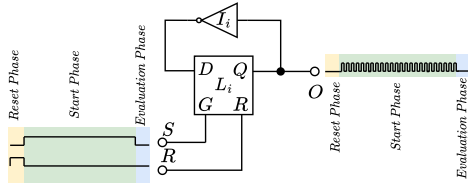


Fig. 1. Scheme of the LRO entropy source.

TABLE I
NIST SP 800-90B ENTROPY TEST RESULT VERSUS SUPPLY
VOLTAGE AND TEMPERATURE VARIATIONS

	V_{DD} [V]			T [deg]		
	0.9	1.0	1.1	5	27	75
H-bitstring	0.997423	0.999679	0.998048	0.993569	0.999679	0.998689
H-original	7.869635	7.892510	7.894144	7.841505	7.892510	7.928804

proposed PUF + TRNG architecture is implemented on the AMD Xilinx Artix-7 FPGA platform. A thorough evaluation of the bitstream behavior has highlighted the robustness of the entropy source also with respect to environmental variations. Measurement on the PUF and TRNG behavior has shown state-of-the-art characteristics for both the PUF and the TRNG configuration, resulting in state-of-the-art performance for reconfigurable entropy source.

II. PROPOSED ARCHITECTURE

A. LRO Cell

The LRO cell is depicted in Fig. 1. The single cell is composed by an inverter and a latch. Two control signals are used in order to properly excite the architecture: S and R . The excitation sequence can be divided in three main phases.

- 1) *Reset Phase*: The R signal is high, whereas the S signal is low. The latch is opaque, and its output is forced to 0. Because of the inverter, the D input of the latch is set to 1, and no oscillation can be triggered at the output node Q .
- 2) *Start Phase*: The R signal goes low, and the S signal is kept high for N clock cycles. Since the S signal controls the gate pin, in this phase, the latch is transparent, and the circuit behaves as an RO, which is left oscillate for N clock cycles.
- 3) *Evaluation Phase*: The R signal stays low, and the S signal goes low too, thus making the latch opaque, and sampling the output Q of the RO in a metastable state. In this phase, both the output of the PUF and the output of the TRNG are generated.

The PUF response can be obtained by comparing the oscillation frequencies of multiple, nominally identical LRO cells, whereas the TRNG bitstream can be extracted by sampling multiple times the LRO oscillation.

B. PUF Model and Working Principle

Since the PUF response is generated by comparing the frequencies of two nominally identical LROs, the entropy source exploited in the PUF is the difference among the periods of the two LROs, which are compared. The period of oscillation of the generic i th LRO is given by three contributions

$$T_{LRO_i} = 2(T_{DQ_i} + T_{INV_i} + T_{R_i}) \quad (1)$$

where T_{DQ_i} is the propagation delay $D \rightarrow Q$ of the latch when the gate pin G is high, T_{INV_i} is the propagation delay of the inverter, and T_{R_i} is the propagation delay of the intraconnections. Each of these propagation delays has two contributions, one related to process (die to die) variations, and the other given by mismatch (within die) variations, $T_i = T_{prc} + T_{mm}$. Since the randomness of the PUF

TABLE II
ENTROPY EXTRACTED FROM THE AIS-31 T8 TEST UNDER
DIFFERENT SUPPLY VOLTAGE AND
TEMPERATURE CONDITIONS

	V_{DD} [V]			T [deg]		
	0.9	1.0	1.1	5	27	75
T8	7.99463	7.99796	7.99897	7.99776	7.99796	7.99947

is computed considering multiple instances on the same chip, the process contribution is common to all the instances of a given chip, and the generic propagation delay can be expressed as $T_i = t_{prc} + T_{mm}$, where T_{mm} is assumed to be a random variable with Gaussian distribution $T_{mm} \sim \mathcal{N}(\mu_i, \sigma_i^2)$.

The difference between the oscillation periods of two LROs can then be expressed as follows:

$$\Delta T = T_2 - T_1 = 2(T_{DQ_2} + T_{INV_2} + T_{R_2} - T_{DQ_1} - T_{INV_1} - T_{R_1}) \quad (2)$$

which, for a given process realization, has the following expression:

$$\Delta T^* = 2 \cdot [T_{mm,DQ_2} + T_{mm,INV_2} + T_{mm,R_2} + - (T_{mm,DQ_1} + T_{mm,INV_1} + T_{mm,R_1})]. \quad (3)$$

Starting from these observations, the PUF response X for a given process realization can be derived as follows:

$$X = \frac{1}{2} - \frac{\text{sgn}\{\Delta T^*\}}{2} \quad (4)$$

where sgn denotes the sign function and $\Delta T^* \sim \mathcal{N}(\mu_{\Delta T^*}, \sigma_{\Delta T^*}^2)$.¹

The expected value of X can be computed as follows:

$$E[X] = \frac{1}{2} - \frac{1}{2} \text{erf}\left(\frac{\mu_{\Delta T^*}}{\sqrt{\sigma_{\Delta T^*}^2}}\right) \quad (5)$$

which tends ideally to 0.5 (Shannon entropy equal to 1), if the two LROs have nominally the same oscillation period.

C. TRNG Model and Working Principle

The TRNG bit extraction comes from the itself sample behavior of the LRO, which occurs N clock cycles after the start of the oscillation. The entropy of the TRNG relies both on jitter accumulation and on the mutual sampling of two oscillators. Timing jitter in the LROs accumulates during the whole start phase (whose duration is $t = N T_{clk}$) with a variance defined as follows [19], [20]:

$$\sigma_t^2 = \kappa^2 \cdot t = \frac{8}{3\eta} \frac{kT}{P} \frac{V_{DD}}{V_{char}} \cdot t \quad (6)$$

where $V_{char} = (V_{DD}/2 - V_{TH})/\gamma$, V_{DD} is the supply voltage, V_{TH} is the threshold voltage of MOS transistors, γ is a technology parameter,² T denotes the absolute temperature, k is Boltzmann's constant, η is a proportionality constant typically close to 1, and the term $P = V_{DD}^2 C_{load} f_{LRO}$ is the power consumption of the RO.³ Denoting with ν the ratio between f_{LRO} and the frequency f_s of the S signal ($\nu = f_{LRO}/f_s$), and with Q the quality factor ($Q = \kappa^2 N T_{clk}$), the probability of sampling a logic 1 after N clock cycles can be written as follows [20]:

$$p = \frac{1}{2} - \frac{2}{\pi} \sin(2\pi\nu) e^{-2\pi^2 Q} + O(e^{-4\pi^2 Q}). \quad (7)$$

¹ $\mu_{\Delta T^*} = \mu_{T_{DQ_2}} + \mu_{T_{INV_2}} + \mu_{T_{R_2}} - \mu_{T_{DQ_1}} - \mu_{T_{INV_1}} - \mu_{T_{R_1}}$ and $\sigma_{\Delta T^*}^2 = \sigma_{T_{DQ_2}}^2 + \sigma_{T_{INV_2}}^2 + \sigma_{T_{R_2}}^2 + \sigma_{T_{DQ_1}}^2 + \sigma_{T_{INV_1}}^2 + \sigma_{T_{R_1}}^2$.

² γ is 2/3 for long channel length and 4/3 or 2 for short channel length.

³ C_{load} is the parasitic load capacitance of the conventional inverter, and f_{LRO} is the LRO oscillation frequency.

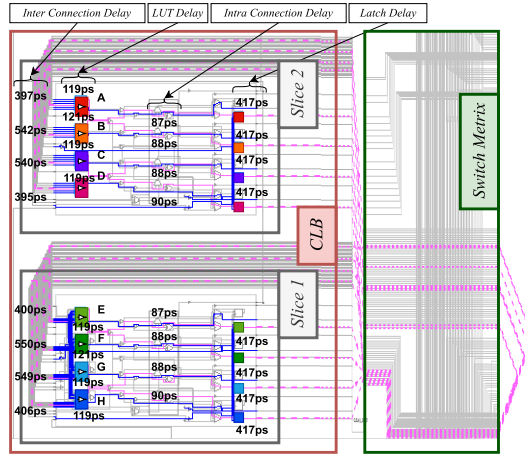


Fig. 2. FPGA implementation of the LRO cells: each LRO cell instance has been highlighted with different colors, the two Slices have been denoted with gray color, and the CLB with red color, whereas the switch matrix, which manages inter-Slice connections, has been highlighted with green color.

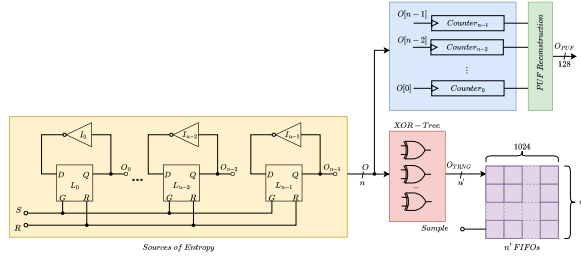


Fig. 3. Scheme of the LRO used as a reconfigurable entropy source for PUF and TRNG output reconstruction.

Equation (7) shows that p tends to the ideal value of 0.5 with the increasing of the number of clock cycles (more jitter is accumulated). The Shannon entropy can be easily computed starting from (7). However, it has to be remarked that, typically, the single LRO cell results in limited entropy, due to the fact that the jitter accumulated in N clock cycles is not enough to guarantee a high quality factor, and an XOR-tree approach like the one presented in [21] can be considered to increase the entropy. In such case, considering an n round XOR-tree architecture, and assuming identical and independent 1-event probability p for each LRO instance, the probability p_{out} of the logic 1 event at the output of the XOR tree can be expressed as follows:

$$p_{out} = \frac{1 - (1 - 2 \cdot p)^{n+1}}{2} \quad (8)$$

which tends to its ideal value by increasing n .

D. Hardware Implementation

In order to implement the LRO cell on FPGA, the AMD series-7 devices have been chosen as target platform. A lookup table (LUT) and a flip-flop, configured as latch, have been exploited to implement the LRO cell. Since each Slice contains 4 LUTs and 4 latches, and it was possible to instantiate 4 LRO cells in each Slice. The configurable logic block (CLB) contains two Slices, and thus, 8 LRO cells have been integrated in each CLB. However, with such dense arrangement, the managing of the inter-Slice connections through the switch matrix is not simple and has to be carefully handled to optimize the performance of the cryptographic primitives. In fact, as can be observed from (5), an asymmetric routing ($\mu_{IR_1} \neq \mu_{IR_2}$) has a strong impact on the bias of the response and, thus, on the key quality. It is clear that without a careful implementation on an FPGA, as meticulously detailed and illustrated in the following,

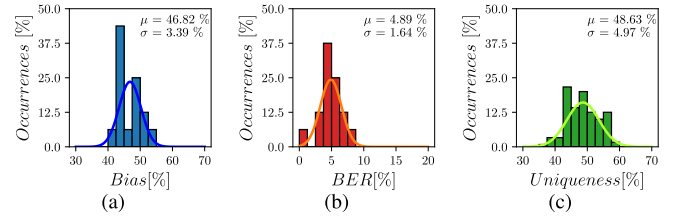


Fig. 4. (a) Bias, (b) BER, and (c) uniqueness of the proposed LRO PUF.

achieving optimal statistical performance is nearly unattainable. As a consequence, to enhance statistical performance, Slice connections have been selected in order to obtain two pairwised LROs, which nominally oscillates at the same frequency, from each Slice. By following this approach, a macroblock containing eight LRO cells has been designed, and its implementation is reported in Fig. 2. Each LRO cell instance has been highlighted with different colors, the two Slices have been denoted with gray color, and the CLB with red color, whereas the switch matrix, which manages inter-Slice connections, has been highlighted with green color. The delays of the intra-Slice and inter-Slice connections, as well as the delays of the latches and of the LUTs, have been reported for each LRO cell within the figure. As it can be observed, LRO cells B-C-F-G (A-D-E-H) oscillates at very similar frequencies (they have very similar delays), and thus, the frequencies of these cells can be compared to extract 8 bits from each CLB.⁴

The architecture proposed in this work to implement both a PUF and a TRNG, starting from several LRO cells exploited as entropy sources, is reported in Fig. 3. The proposed circuit contains 128 identical LRO cells, each activated through the S and R signals, as detailed in Section II-A. The oscillation frequencies of the different LROs are measured using counters (highlighted in blue in Fig. 3), whereas the final state of the oscillation is stored because of the latching behavior of the LRO, as discussed in Section II-A. Following the excitation phase, the TRNG retrieves information from the final state of the oscillation. Conversely, for the PUF, the information is reconstructed by comparing the oscillation frequencies of the nominally identical LRO cells, measured by the counters.

III. TESTBENCH AND MEASUREMENT RESULTS

In order to test the performance of the proposed PUF + TRNG architecture, 16 identical Artix-7 FPGAs (xc7a100t-2e) have been employed. A reference board has been also used to evaluate performance under supply voltage and temperature variations. A programmable power supply unit and a climatic chamber are exploited to vary the operating conditions.

A. PUF Performance Evaluation

Main performances measured for the 128-bit LRO PUF evaluated across 16 FPGA boards are summarized in Fig. 4. The bias of the PUF is reported in Fig. 4(a), and the nominal bit error rate (BER) is shown in Fig. 4(b), and finally, the uniqueness results are reported in Fig. 4(c). Overall, the bias has a mean value of 46.82% (ideal value 50%), the BER has a mean value of 4.89% (ideal value 0%), and the uniqueness distribution shows a mean value of 48.63% (ideal value 50%). These findings highlight the effectiveness of the proposed approach. Furthermore, from bias and uniqueness results, it is clear that the adopted routing strategy and the approach to extract the PUF response have properly addressed the tradeoffs of the FPGA implementation. The reliability (1-BER) has been evaluated with

⁴Frequencies compared among LRO cells coupled as A-D, E-H, B-C, F-G, A-E, B-F, C-G, and D-H.

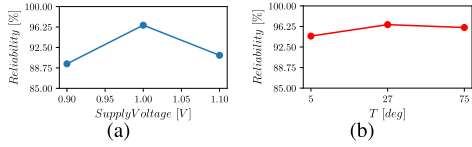


Fig. 5. Reliability of the proposed PUF versus supply voltage variations.

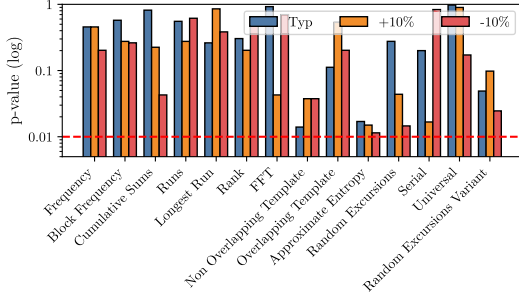


Fig. 6. NIST SP 800-22 graphical representation of p-value results (worst case) with respect to different voltages: the pass rate of each test is always greater or equal than the minimum required to pass NIST tests.

respect to supply voltage variations and depicted in Fig. 5(a), and with respect to temperature variations, and depicted in Fig. 5(b). As it can be observed, temperature variations do not significantly affect the reliability, whereas the supply voltage variations have a relatively high impact on the robustness of the PUF response, resulting in a degradation of the reliability of about 6% (reliability = 90.11%) in the worst case.

B. TRNG Performance Evaluation

In order to evaluate TRNG performance, a preliminary analysis on the standalone entropy source cells has been carried out, with a clock frequency of 75 MHz. The sample signal of the FIFO depicted in Fig. 3 is asserted after the falling edge of the S signal, thus allowing the simultaneous sampling of the 128 bits at the output of the LRO cells. The row bitstream, extracted by collecting the entropy sources outputs without any XOR tree and considering $N = 1$, has been found to exhibit a 1-event probability of 0.542906. In order to reduce the bias of the bitstream, two XOR rounds have been inserted between the entropy sources and the FIFO, as depicted in Fig. 3. The 1-event probability at the output of the XOR tree has then been found to be 0.499961. Considering the two XOR rounds, the final throughput of the proposed TRNG is $TP = 128 \cdot f_{clk} / (2 + N) / 2^{\text{XOR-rounds}} = 128 \times 75 \text{ MHz} / 3 / 2^2 = 800 \text{ Mbit/s}$. In order to implement the two XOR rounds for a 128-bit bitstream, it is necessary to consider 64 additional LUTs and, therefore, 16 additional Slices.

According to [5], a TRNG restart experiment was conducted. The 1000 sequences of 1024 bits were extracted from different start-ups. The Pearson's correlation coefficient follows a normal distribution $\mathcal{N}(2.975 \cdot 10^{-5}, 1.8 \cdot 10^{-4})$, confirming the lack of correlation between the bitstreams. The performance of the proposed TRNG has been evaluated by using the NIST 800-22, the NIST SP 800-90B, and the AIS-31 tests [19]. Exploiting the XOR-tree-based postprocessing with just two rounds, the proposed TRNG is able to pass all the statistical tests (NIST 800-22, NIST SP 800-90B, and AIS-31), also taking into account supply voltage and temperature variations. For the sake of compactness, just the worst case p values for NIST SP 800-22 tests have been reported in Fig. 6, for the typical condition, and under V_{DD} variations. As it can be observed, when the V_{DD} increases of 10%, the randomness improves, whereas when the V_{DD} decreases of 10%, the randomness gets worse. This behavior is in line with (6), in which it can be observed that the σ_r increases with greater V_{DD} . In such condition, the quality factor Q becomes larger, and the second term

TABLE III
COMPARISON TABLE OF PUF + TRNG PERFORMANCES

		Platform	PUF Performance				TRNG Performance				PUF + TRNG
			Slice/bit	Uniq.	Rel.	FOM_{HD}	Slice	TP [Mbit/s]	O.F. [MHz]	FOM_T	
TRNG	[21]	S6	TRNG Only				8	12.5	50	0.028	
	[22]	Z7					33	12.5	125	0.003	
	[23]	A7					4	100	100	0.250	
	[24]	V6					14	100	400	0.018	
	[5]	S6					1	0.76	50	0.016	
PUF	[1]	A7	1	48.05	99.30	48.27	PUF Only				
	[3]	S6	2	49.24	99.18	44.72					
	[4]	S6	1	49.93	93.96	16.56					
	[2]	A7	0.5	49.50	98.62	136.26					
PUF+TRNG	LRO	A7	0.25	48.63	95.11	787.67	48 [‡]	800	75	0.222	174.86
	[13]	A7	0.5	49.48	98.33	114.35	64	225	450	0.008	0.893
	[18]	AF	-	48.82	97.63	-	-	0.9	18	-	-
	[10]	S3	11	39.90	77.90	0.37	322	3.2	-	-	-
	[17]	A7	181	48.47	98.70	0.28	1083	0.024	395	0.06 μ	0.02 μ
	[16]	V6	21	49.79	96.98	1.57	21	100	100	0.048	0.075

A7: Artix-7, AF: Actel-Fusion, S3: Spartan-3, V6: Virtex-6, S6: Spartan-6, Z7: Zynq-7000, [‡] 16 Extra Slice have been considered for the XOR-tree architecture.

of (7) is reduced, improving the bias. For what concerns NIST SP 800-90B and AIS-31 tests, for the sake of compactness, we reported just the entropy per byte, and the entropy per bit, extracted from the NIST SP 800-90B (Table I) and T8 test of AIS-31 (Table II), under the different supply voltage and temperature conditions.

Also, in this case, it can be observed that the entropy decreases with lower supply voltages and temperature, as expected from (6). Furthermore, it has to be remarked that in the worst case, the bit entropy is 0.999328 (extracted at -10% of the nominal supply voltage), which is far beyond the minimum necessary to pass AIS-31 tests, which is 0.997.

IV. COMPARISON WITH STATE OF THE ART

In order to compare the proposed architecture with state-of-the-art PUFs, TRNGs, and PUF + TRNG reconfigurable entropy sources, the FOMs presented in [13] have been used. The FOM_{HD} ,⁵ which mainly evaluate the tradeoff between performance in typical conditions and resources usage in bits/Slice, has been exploited to compare PUF performances. The FOM_T ,⁶ which normalize the TRNG throughput TP to the resources usage and the operating frequency OF, has been adopted to compare TRNGs performance. Finally, the FOM_{Uni} ,⁷ defined as the product between FOM_{HD} and FOM_T , has been exploited to compare the reconfigurable entropy sources with PUF + TRNG behavior. From the comparison reported in Table III, it can be observed how the proposed architecture reaches the minimum resources usage for the PUF topology, and thus, even if the reliability is not as good as other designs, it achieves the best tradeoff between resources consumption and PUF performance. Furthermore, it can also be observed that the proposed LRO-based TRNG achieves the highest TP in the literature (about 3.5 times the state of the art). Finally, for what concerns tradeoffs of the proposed reconfigurable entropy sources, the achieved FOM_{Uni} is about 200 times higher than the state of the art.

V. CONCLUSION

In this work, the LRO has been exploited as a reconfigurable entropy source to implement a PUF + TRNG architecture along with a detailed explanation of the design flow to follow in order to extract robust statistical performance. The proposed LRO-based PUF is able

⁵ $FOM_{HD} = ((\text{bits}/\text{Slice}) / ((0.5 - HD_{inter})^2 + HD_{intra}^2)^{1/2})$.

⁶ $FOM_T = (TP / (OF \cdot \text{Slice}))$.

⁷ $FOM_{Uni} = FOM_T \cdot FOM_{HD}$.

to generate 4 PUF bits per Slice and results in the most compact PUF primitive in the literature. When used as a TRNG, the proposed architecture exhibits the highest throughput in the literature, being able to pass the NIST and AIS-31 tests also under supply voltage and temperature variations. The overall resource consumption of both the reconfigurable entropy source along with the postprocessing technique is 48 Slices. In terms of the tradeoff between PUF and TRNG performance, the proposed reconfigurable entropy source is able to achieve the best FOM_{uni}, which results about 200 times higher than the previous state of the art.

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